

SHIONOMISAKI, Japan

WMO: 477780

Lat: 33.4503N

Lon: 135.7572E

Elev: 227

StdP: 14.58

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	35.4	37.2	17.3	13.3	40.2	19.4	14.7	40.9	27.5	52.8	24.5	51.7	11.1	310	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	87.0	79.0	85.7	78.6	84.5	78.0	80.7	84.7	79.8	83.7	79.1	82.8	10.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.5	154.4	83.2	78.7	150.3	82.5	77.9	146.4	81.9	44.5	84.8	43.6	83.9	42.9	83.1	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.8	21.1	18.7	DB	32.0	89.7	2.3	1.3	30.3	90.6	28.9	91.4	27.6	92.1	25.9	93.1	
			WB	27.0	81.7	2.0	1.0	25.6	82.5	24.4	83.0	23.3	83.6	21.9	84.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.7	46.6	47.8	52.8	60.3	67.2	72.0	78.9	81.0	76.6	68.8	60.0	51.1
	DBStd	12.51	4.79	5.48	5.45	4.78	3.27	3.40	3.08	2.42	3.60	4.16	5.18	5.40
	HDD50	312	127	98	33	1	0	0	0	0	0	0	1	52
	HDD65	2205	570	481	378	152	16	0	0	0	0	12	164	432
	CDD50	5302	22	38	120	309	532	660	894	961	797	583	300	86
	CDD65	1721	0	0	0	10	84	211	429	496	347	131	13	0
	CDH74	11804	0	0	0	2	75	617	3574	4956	2289	284	7	0
	CDH80	2320	0	0	0	0	0	23	655	1287	347	8	0	0
Wind	WSAvg	9.4	9.9	9.7	10.0	9.7	9.5	9.4	9.2	9.2	9.3	9.1	8.6	9.4
Precipitation	PrecAvg	104.2	3.9	4.4	6.9	9.1	10.2	14.7	10.9	10.7	12.4	9.8	7.0	3.9
	PrecMax	142.6	11.5	11.0	14.9	18.7	26.3	23.4	29.3	41.3	33.7	37.6	21.1	12.2
	PrecMin	68.3	0.0	0.7	1.9	3.3	3.9	3.6	1.1	1.4	2.6	1.2	0.8	0.0
	PrecStd	18.2	2.5	2.6	2.8	4.1	3.8	4.7	7.0	6.7	6.4	6.1	5.0	2.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.9	65.9	68.2	73.7	78.1	82.5	88.4	89.1	86.7	81.0	74.6	68.1
		MCWB	59.2	61.7	59.8	64.1	68.5	76.2	78.5	80.0	77.8	73.3	68.2	62.5
	2%	DB	60.2	63.3	66.1	71.2	76.0	80.2	86.3	87.6	84.9	79.0	71.9	64.4
		MCWB	54.2	58.7	60.0	63.6	67.8	74.7	78.8	79.5	77.1	72.0	65.1	57.6
	5%	DB	56.8	60.2	64.3	69.4	74.5	78.6	84.9	86.4	83.4	77.2	69.9	61.7
		MCWB	49.4	54.2	58.2	62.7	66.9	73.7	78.4	79.0	76.3	70.4	63.1	54.5
	10%	DB	54.2	56.9	62.3	67.6	73.0	77.3	83.5	85.3	82.0	75.3	67.9	59.3
		MCWB	46.7	49.9	55.8	60.9	65.8	72.8	77.8	78.5	75.6	68.3	61.3	52.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.0	63.0	64.7	69.3	73.9	78.0	81.0	82.3	80.4	76.9	70.9	64.7
		MCDB	63.0	64.9	66.0	70.6	75.8	80.3	85.1	86.2	84.0	79.1	73.0	67.6
	2%	WB	55.7	59.8	62.0	66.9	71.6	76.8	80.0	81.1	79.0	74.3	67.4	59.3
		MCDB	59.1	62.8	64.6	69.0	73.5	78.7	83.9	85.1	82.7	77.3	70.2	63.0
	5%	WB	51.0	55.5	59.3	64.6	69.5	75.7	79.2	80.3	78.1	72.2	64.8	56.0
		MCDB	54.9	59.0	63.2	67.7	72.3	77.8	82.8	84.4	81.7	75.7	68.4	60.2
	10%	WB	48.0	51.5	56.7	62.5	67.7	74.4	78.6	79.6	77.1	69.9	62.6	53.4
		MCDB	53.1	55.3	61.3	66.4	71.1	76.7	81.9	83.7	80.7	73.9	66.6	58.3
Mean Daily Temperature Range	5% DB	MDBR	10.5	11.5	11.8	10.9	9.6	7.9	7.3	7.9	8.4	9.3	10.2	10.4
		MCDBR	13.3	13.2	12.4	11.6	10.9	8.6	8.7	8.9	9.2	10.1	11.1	12.2
		MCWBR	11.8	12.9	11.7	9.2	7.9	5.5	4.6	4.8	5.5	7.7	9.6	10.7
	5% WB	MCDBR	12.3	12.9	11.8	9.7	8.7	7.0	7.6	8.0	8.1	8.8	10.4	11.4
		MCWBR	13.2	14.0	13.1	9.6	7.9	5.1	4.4	4.5	5.5	8.2	10.7	11.9
Clear-Sky Solar Irradiance	taub	0.393	0.440	0.500	0.532	0.540	0.547	0.545	0.513	0.487	0.465	0.420	0.387	
	taud	2.210	2.091	1.941	1.911	1.967	2.037	2.081	2.173	2.210	2.180	2.218	2.244	
	Ebn at Noon	252	252	246	245	244	241	241	247	248	242	241	247	
	Edh at Noon	36	45	57	61	58	54	52	47	43	41	36	33	
All-Sky Solar Radiation	RadAvg	838	1037	1350	1556	1650	1416	1705	1755	1368	1056	861	762	
	RadStd	66	83	95	174	185	151	192	136	112	112	88	46	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page